CRED-RAPTOR: A Hierarchical RAG-based LLM Pipeline for Corporate Credit Risk Assessment (Extended Abstract)

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Abstract

Corporate credit risk assessment is a crucial activity for banks as it helps to predict the default risk associated with corporate borrowers. Banking institutions face significant challenges in auditability, manual effort, and the inefficient use of unstructured data. Traditional assessment methodologies rely on manual analysis by risk analysts who evaluate financial data alongside various internal documents, such as credit memos, to calculate credit scores for each company. This approach lacks transparency and fails to leverage the full capabilities of large documents.

This study builds an LLM pipeline for credit assessment, employing a hierarchical Retrieval-Augmented Generation framework to extract qualitative features like business insights, operational risks, revenue stability, regulatory compliance, and management execution from the bank's lengthy unstructured documents, which are often overlooked during manual credit assessment. These qualitative features are then incorporated into a structured format alongside traditional quantitative financial metrics to train a machine learning model for predicting a company's credit score. The pipeline also addresses cold-start companies with no prior institutional history by leveraging data from their peers. Moreover, our approach addresses explainability and transparency in the credit assessment process by depicting the exact features used to generate credit scores, which helps with audits.

1 Introduction

Corporate credit risk assessment is a critical and fundamental process in banking, enabling institutions to evaluate the default probability of corporate borrowers through systematic analysis of financial and non-financial features. Traditional frameworks predominantly rely on quantitative metrics such as leverage ratios, profitability measures, and liquidity indicators to predict corporate performance and creditworthiness. However, research increasingly demonstrates that quantitative analysis alone misses several angles of corporate creditworthiness that the qualitative features can reveal.

Soares et al. [1] surveyed loan officers across six major Portuguese banks and found that qualitative factors like management experience and reliability were the most valued factors in credit decision making. Their study also revealed a strong negative correlation between a bank's emphasis on qualitative features and overdue credit performance, demonstrating that the importance of qualitative features alongside quantitative features improves credit risk outcomes. In recent years, AI and machine learning integration has shown substantial promise, with studies reporting 30% and 40% improvements in prediction accuracy and reduction in default rates, respectively [2]. However, the "black-box"model creates interpretability problems conflicting with regulatory requirements for explainable credit decisions. Here, explainability refers to the transparency of which features are extracted and how they influence the outcome, not the model's internal scoring. The black-box model is used solely for feature extraction from lengthy documents, with interpretable methods applied thereafter. Retrieval-Augmented Generation (RAG) frameworks further ensure traceability by grounding extracted features in verifiable source documents, supporting the auditability of credit decisions [3]. Advanced RAG architectures such as RAPTOR build a hierarchical tree of document summaries through recursive clustering, enabling retrieval at multiple levels of abstraction and demonstrating stronger performance on knowledge-intensive tasks compared to plain retrieval approaches [4]. Also, hybrid systems combining RAG-based extraction with predictive models ensure the predictions are supported by retrieved information [5]. This study implements an integrated pipeline combining a hierarchical RAG with an LLM for qualitative feature extraction, and uses XGBoost for credit score prediction, including a novel clustering-based peer approach for cold-start companies.

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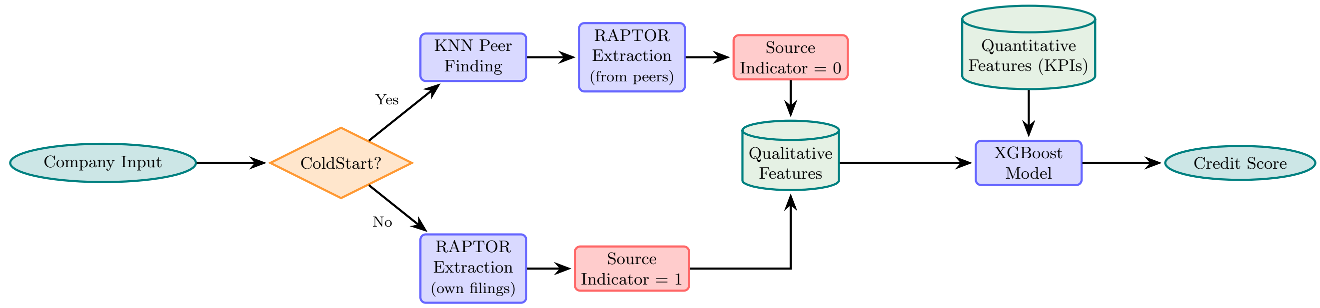

Figure 1: Corporate Credit Risk Assessment Pipeline

2 Proposed Methodology

This study proposes a hybrid LLM and machine learning pipeline developed for credit score prediction, drawing on three distinct feature sets. Quantitative and qualitative features are sourced from US SEC 10-K filings, complemented by a binary indicator that distinguishes companies with existing filings from cold-start companies without prior institutional records.

Quantitative Feature Calculation. Three financial KPIs are calculated programmatically from the US SEC filings dataset.

1. Profit Margin (net income / net sales × 100) measures operational efficiency.
2. Return on Assets (net income / total assets × 100) assesses how effectively firms utilized their assets to generate earnings.
3. Debt Ratio (long-term debt / total assets × 100) captures financial leverage.

All indicators are rounded to two decimal places and computed only when both required variables are present and the denominator is non-zero.

Qualitative Feature Extraction. Company 10-K filings are processed using RAPTOR (Recursive Abstractive Processing for Tree-organized Retrieval) [5], which overcomes the limitations of chunking approaches by capturing both granular details and broad thematic context simultaneously. When the long document is ingested, it gets segmented into contiguous chunks, embedded using SBERT, then softly clustered using Gaussian Mixture Models, allowing chunks to belong to multiple clusters simultaneously. Each cluster is summarized by an LLM into a parent node, which is re-embedded and re-clustered recursively, building a tree bottom-up until no further clustering is possible. Nodes are retrieved using cosine similarity until a token threshold is met. The retrieved context is then passed to an LLM prompt, scoring five credit risk dimensions - industry and business risk, revenue stability risk, operational and asset risk, Management and execution risk, and regulatory and licensing risk - on a ten-point scale. Neutral scores were assigned as fallback values when the output was ambiguous or malformed.

Source Indicator. A binary feature, the source indicator, depicts whether the company is a cold start or has existing filings (US SEC filings) used in extracting qualitative features.

Credit Score Prediction. All three features are then trained with an XGBoost model to predict a credit score of a corporate, ranging from 0 to 10.

Cold Start Companies. For cold-start companies, K-nearest-neighbour clustering is proposed to identify peers and leverage the qualitative features of those peers.

3 Future Work and Scope

The model utilizes US SEC filings data, which closely resembles a credit memo. Fine-tuned LLMs with domain expertise could be employed to improve qualitative assessment. Since ground-truth credit scores are not publicly available in SEC filings, LLMs could also be used to synthetically synthesize credit score labels for training purposes. For cold-start companies specifically, where input financial data is sparse, data augmentation could be applied to generate additional training samples. This would enrich both the financial profiles and their corresponding credit score labels making the training set more complete and representative. Additionally, introducing caching in various pipeline stages could improve the auditability and traceability of any credit decision made to satisfy the explainability requirements.

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